

Dear Search Committee,

I am writing to apply for the Assistant Professor Position (Ref. AC2026/011/01) in the School of Artificial Intelligence (SAI) at The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen). I am currently a Research Fellow at Nanyang Technological University (NTU), Singapore, where I received my Ph.D. in Robotics and Intelligent Systems. My research focuses on embodied intelligence and AI foundations for autonomous systems, particularly multi-agent reinforcement learning (MARL), vision-based learning, and scalable learning frameworks for robots operating in complex, dynamic, and adversarial environments. I am excited by SAI's mission to address national strategic needs while advancing international AI frontiers, and I am eager to contribute to its interdisciplinary, internationalized teaching and research ecosystem in Shenzhen and the Greater Bay Area.

My research advances AI methodologies that enable robust, safe, and scalable autonomy in the real world. During my Ph.D. and postdoctoral work, I developed learning-based control and multi-agent decision-making for swarms of aerial and ground robots in tasks such as multi-target search and rescue, pursuit-evasion, and operations under uncertainty and adversarial disturbances. These projects are not only algorithmic: they integrate perception, dynamics, and control with full-stack deployment, including onboard sensing, real-time policy execution, and field experimentation. In parallel, I have worked on risk-aware decision-making for autonomous navigation using reinforcement learning with transformer-based perception, and on learning-enhanced prediction in physics-governed systems, building a consistent theme of combining data-driven intelligence with structure, safety, and physical constraints.

At CUHK-Shenzhen SAI, I plan to build a research program aligned with the School's focus on AI foundations, embodied AI, large-scale learning, and robotics/autonomous systems. My near-to-mid-term agenda has three directions:

(1) Trustworthy Multi-Agent Learning for Dynamic Environments. I will develop MARL methods that are robust to non-stationarity, partial observability, and adversarial perturbations, with explicit treatment of safety, uncertainty, and accountability. This includes safe policy learning with constraint satisfaction, reliability-aware coordination, and interpretable decision modules, supporting applications from swarm robotics to intelligent navigation and human-in-the-loop autonomy.

(2) Embodied AI with Multi-Modal Perception and Foundation Models. I will explore vision-language-action policies and world-model-based learning that connect multi-modal perception (vision, state, maps, language) to planning and control. A key goal is improving generalization and sim-to-real transfer by incorporating structured priors (e.g., neuro-symbolic or cognitive-inspired elements) and by designing evaluation benchmarks that reflect real operational constraints.

(3) Large-Scale and Distributed Learning for Multi-Agent Systems. I will develop scalable training and deployment pipelines for multi-agent learning distributed simulation, efficient data collection, and on-device adaptation, to enable practical multi-robot learning frameworks. This direction naturally connects to Shenzhen's strong industrial ecosystem in robotics, edge computing, and AI systems, enabling impactful academia-industry collaborations and translational outcomes.

I believe SAI's strategic location and interdisciplinary foundation create an ideal environment for this program. I am particularly interested in initiating collaborations across CUHK-Shenzhen and Shenzhen/GBA partners on embodied autonomy, scalable learning systems, and safety-critical AI, and in pursuing competitive funding opportunities (e.g., NSFC, Guangdong/Shenzhen programs, and joint industry-supported projects) to sustain a high-impact, externally funded research group.

Teaching and mentorship are central to my academic vision. At NTU, I have contributed to course development and teaching in autonomous flight, robotics, and artificial intelligence, including hands-on tutorials and project supervision. If given the opportunity to join SAI, I would be enthusiastic to teach both core and advanced courses such as Reinforcement Learning, Multi-Agent Systems, Machine Learning Foundations, Motion Planning and Intelligent Navigation, Embodied AI, and Applied AI for Autonomous Systems. My teaching style emphasizes strong fundamentals (math, optimization, learning theory) paired with system-level thinking and implementation, helping students connect algorithms to real deployment constraints through well-scaffolded projects.

Beyond research and teaching, I am committed to academic service through reviewing, community engagement, open-source release of research artifacts when appropriate, and student mentorship. I would be honored to contribute to SAI's growth as a newly established school by helping build curriculum, labs, interdisciplinary collaborations, and a vibrant research culture with strong international visibility.

Thank you for considering my application. Look forward to the opportunity to discuss how my research and teaching plans align with the School of Artificial Intelligence at CUHK-Shenzhen and how I can contribute to SAI's mission in Shenzhen and the Greater Bay Area.

Yours sincerely,

Jiaping Xiao

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